MITRE ATT&CK Abilities
Expanded User Manual

Deep explanation of every entry in abilities.json, safe execution guidance, Wazuh/Sysmon monitoring, and Tailscale setup on Kali Linux

Source: your uploaded abilities.json

Audience: learner / blue-team lab operator

Safety boundary: this file includes entries that can expose credential material, create accounts, establish persistence, or refer to remote payload execution. This guide explains those entries and how to detect them, but does not give runnable steps for dangerous actions. Use an isolated, authorized lab only.

1. How to read the tool configuration

The supplied file is JSON data. Each key (for example, T1033) identifies an ability entry, not an executable program by itself. A separate runner must load the JSON and decide whether/how to invoke the value in cmd. Opening or validating the JSON does not run those commands.

Field	Meaning	Review question
Technique key	ATT&CK-style technique identifier or project-specific identifier.	Does the ID match the behavior actually performed?
name	Display label for the ability.	Does the label accurately describe the command?
cmd	Command text stored by the tool.	Could it change the system, expose data, or contact another host?
tactic	High-level objective category.	Is the tactic consistent with observed behavior?
severity	Project-defined severity label.	What local rubric assigns this severity? It is not an official ATT&CK score.

Important: technique mappings in the file should be independently verified against the current MITRE ATT&CK knowledge base. A command can be a read-only check related to a technique rather than an implementation of that technique.

2. Kali Linux: Tailscale installation and use

Tailscale is a mesh VPN based on WireGuard. It can connect your own Kali machine, Windows test endpoint, and other authorized devices through a private tailnet. It is a networking/access tool, not a MITRE ATT&CK executor. Do not expose a test service to the internet merely because Tailscale is installed.

2.1 Install on Kali

Use the official Tailscale Linux installation instructions for the current supported method and your Kali/Debian base. On a typical Kali system, update package metadata first:

```
sudo apt update
```

Then follow the current official Linux instructions at https://tailscale.com/kb/1031/install-linux. Package/repository steps can change; use the commands shown for Debian-based distributions rather than copying an old third-party script.

2.2 Start the service and authenticate

```
sudo systemctl enable --now tailscaled
sudo tailscale up
```

What these do:

• **systemctl enable --now tailscaled** starts the Tailscale daemon now and enables it at boot. If the package uses a different service state on your system, check it with the status command below.

• **tailscale up** initiates sign-in and device enrollment. Follow the displayed authentication URL and approve only the device/account you control.

```
systemctl status tailscaled --no-pager
tailscale status
tailscale ip -4
```

The status command shows tailnet peers and connectivity; the IP command shows this machine's Tailscale IPv4 address. Treat tailnet addresses and peer names as sensitive lab information.

2.3 Connect to your Windows lab safely

Install Tailscale on the Windows test VM using the official Windows client, sign in to the same authorized tailnet, and verify that both devices appear in the status output. Use the Tailscale IP for a test connection only when you have configured a specific service and firewall rule. Tailscale does not automatically make every port accessible; local firewalls and service binding still apply.

2.4 Troubleshooting / removal

```
sudo systemctl restart tailscaled
sudo journalctl -u tailscaled --no-pager -n 50
```

If the device cannot connect, check service status, authentication, tailnet policy, local firewall, and whether both endpoints are online. Do not disable the firewall wholesale.

To disconnect this device from the tailnet, use `sudo tailscale down`. To remove the software, use your package manager after reviewing the package name installed on your system. Removing a client does not replace revoking a lost device in the Tailscale admin console.

Official reference: Tailscale Linux installation and Tailscale CLI reference.

3. Safe workflow before running any ability

Step 1 — Isolate

Use a disposable Windows VM, snapshot it, and use synthetic files/accounts. Keep the lab separate from production and personal data.

Step 2 — Preserve and validate

Keep the original abilities.json unchanged. Validate syntax with the read-only command below; this does not execute the command strings.

Step 3 — Review the runner

Find where the application reads the JSON and whether it executes cmd values, uses administrator rights, interpolates placeholders, downloads files, or runs commands automatically. Do not run an unknown runner.

Step 4 — Enable telemetry

Confirm Wazuh agent connectivity and collection of Windows Security, Sysmon, and PowerShell logs before testing.

Step 5 — Test one benign item

Choose one read-only inventory ability. Record the host, account, start time, expected behavior, and event evidence.

Step 6 — Review and reset

Search raw events and alerts, document detection gaps, then revert the VM snapshot and check for residual changes.

Read-only JSON syntax validation

```
python -c "import json; json.load(open('abilities.json', encoding='utf-8')); print('JSON syntax OK')"
```

Run this from the directory containing abilities.json. It checks syntax only, not safety, technique accuracy, or the behavior of any external runner.

4. Every ability in the supplied file

The table below covers every key in your source. 'Read-only' means the stored command appears to inspect/list information; still use an authorized lab because output may contain sensitive information. Risk labels are those supplied by your JSON.

T1033 — System Owner/User Discovery

Tactic in source: Discovery **Severity in source:** low

Shows the current user and token/group privileges. Useful for understanding which account and permissions a process has. Output can reveal group membership.

Command from source:

```
whoami /all
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1087 — Account Discovery

Tactic in source: Discovery **Severity in source:** low

Lists local user accounts and local Administrators group members. Useful for account inventory; do not assume it discovers domain accounts comprehensively.

Command from source:

```
net user; net localgroup administrators
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1016 — Network Configuration Discovery

Tactic in source: Discovery **Severity in source:** low

Displays IP settings and routing table. Helps establish interface addresses, gateways, DNS configuration, and routes.

Command from source:

```
ipconfig /all; route print
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1057 — Process Discovery

Tactic in source: Discovery **Severity in source:** low

Lists processes ordered by CPU use. Process names, IDs and paths help baseline endpoint activity; process listings alone do not prove maliciousness.

Command from source:

```
Get-Process | Sort-Object CPU -Descending | Select-Object -First 15 Name,Id,CPU,Path
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1082 — System Information Discovery

Tactic in source: Discovery **Severity in source:** low

Displays Windows system information such as OS/build and hardware details. Useful for inventory and confirming the test host.

Command from source:

```
systeminfo
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1049 — Network Connections Discovery

Tactic in source: Discovery **Severity in source:** low

Filters network connection output for established connections. Connection ownership and remote endpoints should be correlated with process IDs and expected applications.

Command from source:

```
netstat -ano | Select-String 'ESTABLISHED'
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1018 — Remote System Discovery

Tactic in source: Discovery **Severity in source:** medium

The stored command is `arp -a`, which shows the local ARP cache. This is not a full remote-host discovery scan; it only shows recently learned local-network mappings.

Command from source:

```
arp -a
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1083 — File and Directory Discovery

Tactic in source: Discovery **Severity in source:** medium

Lists directory names immediately under C:\Users. It does not recursively read all files; usernames and folder names can still be personal data.

Command from source:

```
Get-ChildItem C:\Users -Directory | Select-Object Name
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1059.001 — PowerShell Execution

Tactic in source: Execution **Severity in source:** medium

Displays PowerShell version and execution policy. It does not change policy or execute a downloaded script. Execution policy is not a complete security boundary.

Command from source:

```
$PSVersionTable; Get-ExecutionPolicy -List
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1012 — Query Registry

Tactic in source: Discovery **Severity in source:** medium

Reads a current-user/machine Run-key location. Registry Run keys may identify programs configured to start at logon; the command uses HKLM, so its name/description should be reviewed carefully.

Command from source:

```
Get-ItemProperty -Path 'HKLM:\Software\Microsoft\Windows\CurrentVersion\Run' -ErrorAction SilentlyContinue
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1518 — Software Discovery

Tactic in source: Discovery **Severity in source:** low

Reads uninstall registry entries and lists software names/versions. Inventory may help identify installed applications, but output depends on registry view and installer type.

Command from source:

```
Get-ItemProperty HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\* | Where-Object { $_.DisplayName } | Select-Object
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1124 — System Time Discovery

Tactic in source: Discovery **Severity in source:** low

Displays local date/time and time zone. Correct time is important when correlating endpoint events with Wazuh alerts.

Command from source:

```
Get-Date; Get-TimeZone
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1552.001 — Credentials In Files

Tactic in source: Credential Access **Severity in source:** high

Searches user files for the word 'password'. This can expose private data or secrets and may generate substantial file-system activity. Use fake test files only; do not scan real user documents.

Command from source:

```
Get-ChildItem C:\Users -Recurse -Include *.txt,*.xml,*.config,*.ini -ErrorAction SilentlyContinue | Select-String -Pattern 'pa
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1555 — Credentials from Password Stores

Tactic in source: Credential Access **Severity in source:** high

Lists saved credential targets via Windows Credential Manager. This can expose sensitive account metadata; avoid collecting or sharing real output.

Command from source:

```
cmdkey /list
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1053.005 — Scheduled Task

Tactic in source: Persistence **Severity in source:** high

Lists enabled/non-disabled scheduled tasks. Enumeration is read-only; task names, actions, triggers, and principals can be useful persistence investigation evidence.

Command from source:

```
Get-ScheduledTask | Where-Object { $_.State -ne 'Disabled' } | Select-Object TaskName -First 15
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1547.001 — Registry Run Keys

Tactic in source: Persistence **Severity in source:** high

Reads the current user's Run key. It inspects auto-start configuration and does not itself create a persistence entry.

Command from source:

```
Get-ItemProperty 'HKCU:\Software\Microsoft\Windows\CurrentVersion\Run' -ErrorAction SilentlyContinue
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1562.001 — Disable Security Tools (Detection Check)

Tactic in source: Defense Evasion **Severity in source:** critical

Despite the label 'Disable Security Tools', the stored command queries Microsoft Defender status fields; it does not disable Defender. Treat this as a security-health check and verify the label against the actual behavior.

Command from source:

```
Get-MpComputerStatus | Select-Object AMServiceEnabled,RealTimeProtectionEnabled,AntivirusEnabled,BehaviorMonitorEnabled,IsTamp
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1005 — Data from Local System

Tactic in source: Collection **Severity in source:** high

Enumerates a limited number of files under Desktop/Documents and outputs names/lengths. File names and metadata can be sensitive; use synthetic files.

Command from source:

```
Get-ChildItem $env:USERPROFILE\Desktop,$env:USERPROFILE\Documents -Recurse -File -ErrorAction SilentlyContinue | Select-Object
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1490 — Inhibit System Recovery

Tactic in source: Impact **Severity in source:** critical

The stored command lists shadow-copy information. It does not delete shadow copies or inhibit recovery. The technique label describes a broader behavior than this read-only command.

Command from source:

```
vssadmin list shadows
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1105 — Ingress Tool Transfer

Tactic in source: Command and Control **Severity in source:** high

Placeholder that prints a message indicating a payload is supplied at deployment time. Inspect the runner/deployment path; never add an untrusted payload or URL.

Source command: placeholder only; it prints a message and does not itself transfer a tool.

```
Write-Host 'T1105 default (payload injected at deploy time)'
```

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1003.002 — OS Credential Dumping - SAM

Tactic in source: Credential Access **Severity in source:** critical

The source command attempts to save SAM, SYSTEM, and SECURITY registry hives to a temporary directory. These hives contain sensitive authentication material. Do not execute it; validate detections through approved simulation or synthetic telemetry.

Source command: intentionally not reproduced as a runnable command because it exports credential-related data, creates privileged access, or establishes remote persistence. Refer to the original JSON for controlled review.

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1003.004 — OS Credential Dumping - LSA Secrets

Tactic in source: Credential Access **Severity in source:** critical

The source command attempts to save the SECURITY registry hive. This is sensitive credential-related material. Do not execute it; use safe detection tests instead.

Source command: intentionally not reproduced as a runnable command because it exports credential-related data, creates privileged access, or establishes remote persistence. Refer to the original JSON for controlled review.

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1078.001 — Create Local Admin Account

Tactic in source: Persistence **Severity in source:** critical

The source command creates a local account with a hard-coded password and adds it to Administrators. This changes the system and can create unauthorized privileged access. Do not execute it; use a formally approved benign account test and remove it through documented cleanup.

Source command: intentionally not reproduced as a runnable command because it exports credential-related data, creates privileged access, or establishes remote persistence. Refer to the original JSON for controlled review.

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

T1543.003 — Windows Service Enumeration

Tactic in source: Persistence **Severity in source:** high

Lists running Windows services. Despite the service-related technique label, the command shown does not create or modify a service; it is inventory only.

Command from source:

```
Get-Service | Where-Object { $_.Status -eq 'Running' } | Select-Object -First 15 Name,DisplayName,Status
```

Basic execution: on the authorized Windows lab endpoint, open PowerShell or Command Prompt as appropriate, paste the displayed read-only command, press Enter, and review output locally. Avoid elevated privileges unless genuinely required.

T1053.005.system — Scheduled Task Persistence (SYSTEM)

Tactic in source: Persistence **Severity in source:** critical

The source command creates a SYSTEM-level logon scheduled task that references hidden PowerShell and a remote payload URL. This can establish privileged persistence and remote execution. Do not execute it; inspect it as code and use safe simulation/synthetic events for detection validation.

Source command: intentionally not reproduced as a runnable command because it exports credential-related data, creates privileged access, or establishes remote persistence. Refer to the original JSON for controlled review.

Testing guidance: do not run this entry directly on a live or personal endpoint. Use synthetic data, approved simulations, or existing endpoint telemetry to validate detection.

5. Wazuh + Sysmon: step-by-step detection workflow

1 — Confirm the Windows agent

In the Wazuh dashboard, confirm the Windows endpoint is active and reporting. Fix agent connectivity before generating test activity.

2 — Confirm log sources

Verify Sysmon operational events and Windows Security / PowerShell logs are being collected. Process creation telemetry is useful for command-line analysis.

3 — Record baseline

Record hostname, user, time zone, time window, and normal behavior. Ensure Kali and Windows clocks are reasonably synchronized.

4 — Run one benign ability

Choose a read-only inventory check from Section 4. Do not batch-run the whole JSON.

5 — Search raw events first

Filter by endpoint and time range, then inspect process image, parent process, user, command line, event source and event ID. A missing alert may mean no matching rule, not no telemetry.

6 — Map observed behavior

Map the event evidence to ATT&CK; based on what actually happened. Do not rely only on the ability key or display label.

7 — Document and reset

Record event/alert IDs, detection result, false positives, gaps, and remediation. Revert the lab snapshot after the exercise.

Test record field	What to capture
Test ID	Unique label, e.g. LAB-DISC-001
Host / user	Endpoint and authorized test account
Technique	Observed behavior plus verified ATT&CK mapping
Time	Start/end timestamps and timezone
Evidence	Event source, event ID, process tree, alert/rule ID
Outcome	Alerted / telemetry only / not observed
Cleanup	Snapshot reverted and residual changes checked

6. Risk classification and operational notes

Group	Examples from this file	Recommended handling
Low-impact inventory	T1033, T1016, T1057, T1082, T1124	Run individually in an authorized lab; review outputs for sensitive details.
Potentially sensitive discovery	T1087, T1083, T1012, T1518, T1552.001, T1555, T1005	Use synthetic data and minimize collection; avoid saving real secrets or personal data.
Security / recovery checks	T1562.001, T1490	The listed commands are checks; they do not perform the destructive action implied by some labels.
Credential hive export	T1003.002, T1003.004	Do not execute. Validate alerts using safe simulation and approved detection testing.
Account / persistence / remote execution	T1078.001, T1053.005.system	Do not execute. Review as code; use harmless simulation or synthetic telemetry.

Severity in the JSON is project metadata and should not be confused with MITRE ATT&CK severity (ATT&CK does not assign a universal severity score to each technique). Define your own severity rubric based on impact, privileges, data sensitivity, and lab

scope.

7. Troubleshooting

JSON syntax error

Check commas, quotes, braces, and backslash escaping. Use the syntax-validation command in Section 3.

PowerShell command fails

Confirm the shell is PowerShell, not cmd.exe. Check Windows version and permissions. Do not bypass policy or endpoint controls to force it to run.

No Wazuh alert

Check agent status, log collection configuration, event timestamps, and raw events. Detection rules do not cover every behavior by default.

Tailscale device is offline

Check `systemctl status tailscaled`, authentication, tailnet policy, connectivity, and local firewall. Review `journalctl` logs.

Technique ID and behavior disagree

Use the current official ATT&CK; knowledge base to verify the mapping. Record the mismatch rather than silently assuming the JSON label is correct.

Unknown runner behavior

Review how it parses and invokes `cmd`, whether it runs as administrator, and whether it substitutes `__C2__` or deploy-time payloads. Do not launch it until understood.

8. Final checklist

- ■ Lab isolated and snapshot available
- ■ abilities.json preserved and syntax checked
- ■ Runner reviewed before use
- ■ Tailscale installed from official instructions and only authorized devices enrolled
- ■ Wazuh agent, Sysmon, and required Windows logs verified
- ■ One benign ability tested at a time
- ■ Sensitive/persistence entries excluded from direct execution
- ■ Raw events and alerts reviewed and evidence documented
- ■ ATT&CK; mappings verified against actual observed behavior
- ■ VM reverted and cleanup confirmed

References: supplied abilities.json; Tailscale Linux installation; Tailscale CLI reference; official MITRE ATT&CK knowledge base at attack.mitre.org.